

Workshop Guide

De Novo Protein Binder Design with RFdiffusion3

RFdiffusion3 >> LigandMPNN / solMPNN >> ColabFold (AF2) >> Screening

Companion material for the Google Colab notebook

Workshop_RFD3_Colab_v4_fixed_v2

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1. Introduction & Pipeline Overview

This workshop walks you through an end-to-end computational protein design pipeline running entirely on Google Colab with a GPU runtime. Starting from a target protein structure (PDB file), you will generate novel binder proteins that are predicted to interact with the target at user-defined hotspot residues.

The pipeline chains four tools, each solving one piece of the design problem:

Stage	Tool	Input	Output
1 - Backbone	RFdiffusion3	Target PDB + contig	Backbone coords (.cif.gz)
2 - Sequence	LigandMPNN / soIMPNN	Backbone coords	Amino acid seqs (.fasta)
3 - Validation	ColabFold AF2 v3	Complex sequences	Pred. structures (.pdb)
4 - Screening	Custom Python	Scores + structures	Metrics CSV + plots

Why this order? RFdiffusion3 operates in 3D coordinate space - it generates the shape of a protein but has no knowledge of sequence. LigandMPNN (or soIMPNN) solves the inverse-folding problem: given a backbone shape, what amino acid sequence will fold into it? Finally, ColabFold re-predicts the complex from sequence alone. If the predicted structure matches the designed backbone, the design is self-consistent and a strong candidate for experimental validation.

Estimated runtime: 30-90 minutes on a T4 or A100 GPU, depending on the number of backbones and sequences you generate.

2. Prerequisites & Environment Setup

Before running any design cells, the notebook installs all required software. Below is an overview of what each setup cell does and why.

Cell 0.1 - GPU Check & Core Packages

Verifies that a GPU is available (nvidia-smi) and installs core Python packages: biopython (PDB parsing), py3Dmol (3D visualization), pandas, numpy, matplotlib (data analysis and plotting).

IMPORTANT: Before running, go to Runtime > Change runtime type > GPU (T4 or A100). The pipeline cannot run on a CPU-only runtime.

Cell 0.2 - Install Foundry (RFdiffusion3 + MPNN)

Installs the official RosettaCommons 'rc-foundry' package from PyPI. Foundry bundles both the rfd3 and mpnn command-line tools. This replaces older manual installs from GitHub that were fragile and often broke.

Cell 0.3 - Install ColabFold + JAX + Relax Dependencies

Installs ColabFold (AlphaFold2 Multimer) with CUDA-enabled JAX for GPU-accelerated structure prediction. Also installs openmm and pdbfixer, which are required by ColabFold's Amber relaxation feature (use_relax = True). Without these two packages, enabling relaxation will fail with a ModuleNotFoundError for pdbfixer.

```
# ColabFold + structure prediction
pip install colabfold[alphafold-minus-jax]

# Amber relax dependencies (required for use_relax = True)
pip install openmm pdbfixer

# GPU-accelerated JAX
```

```
pip install --upgrade jax[cuda12]
```

A compatibility fix cell (0.3b) pins JAX to < 0.11 because ColabFold's dm-haiku dependency relies on an older JAX internal API.

If ColabFold fails with 'DropVar' or 'jax.core' errors, re-run cell 0.3b to pin JAX correctly.

3. Stage 0: Configuration & Target Protein

Cell 1.0 - Design Configuration

This is the central configuration cell where you define all design parameters. Every parameter is exposed as a Colab form widget.

Parameter	Type	Default	Description
TARGET_NAME	string	FPR2_v2	Short name for the target protein
TARGET_PDB	string	FPR2.pdb	Filename of the target PDB file
TARGET_CHAIN	dropdown	B	Chain ID in the PDB to design against
BINDER_MIN	slider	5	Minimum binder length (residues)
BINDER_MAX	slider	20	Maximum binder length (residues)
HOTSPOTS_STR	string	84,88,...	Comma-separated hotspot residue numbers
NUM_BACKBONES	slider	4	Number of backbone structures to generate
MPNN_SEQS	slider	4	Sequences per backbone
MPNN_MODEL	dropdown	ligandmpnn	Inverse-folding model (ligandmpnn or solmpnn)

The total number of designs = NUM_BACKBONES x MPNN_SEQS. For a quick test, 4 x 4 = 16 designs is a good starting point. For a production run, 20-50 backbones x 8-16 seqs (160-800 designs) is typical.

The Contig String

RFdiffusion3 uses a 'contig string' to specify what to generate. The notebook builds it automatically from your parameters:

```
CONTIG = f"{BINDER_MIN}-{BINDER_MAX},/0,{TARGET_CHAIN}1-{len(TARGET_SEQUENCE)}"
```

This means: 'Design a binder of BINDER_MIN to BINDER_MAX residues, then a chain break (/0), followed by all residues of TARGET_CHAIN.' The chain break tells RFdiffusion3 that binder and target are separate protein chains.

Cell 1.1 - Load Target PDB

Two options: upload from your computer or download from RCSB by PDB ID. The cell extracts the target sequence from the specified chain and updates the contig string with the actual sequence length.

Cell 1.2 - Download Model Checkpoints

Downloads pre-trained weights for RFdiffusion3, LigandMPNN, and SolubleMPNN (~600 MB total) using 'foundry install'. Checkpoints are stored in /content/checkpoints and located automatically.

4. Stage 1: RFdiffusion3 - Backbone Generation

RFdiffusion3 (RFD3) uses a denoising diffusion process to generate novel protein backbone structures. Starting from random 3D noise, it iteratively refines atomic coordinates, conditioned on:

- The target protein structure (your PDB)
- The contig string (binder length + target chain)
- Hotspot residues (where the binder should make contact)

How Diffusion Works (Simplified)

1. Start with random atom positions (pure noise)
2. A neural network predicts 'what does the real structure look like?'
3. Subtract a small amount of noise based on the prediction
4. Repeat steps 2-3 many times until a clean backbone emerges

The network was trained on thousands of real protein structures and has learned what makes a physically plausible protein fold. Hotspot conditioning ensures the generated binder makes contacts with your residues of interest.

Hotspot Residues

Hotspot residues are positions on the target chain where you want the binder to make contact. The notation in the notebook maps each residue to a 'CA' (alpha-carbon) atom:

```
SELECT_HOTSPOTS = {"B84": "CA", "B88": "CA", "B172": "CA", ...}
```

Choose hotspots based on:

- Known functional sites (active sites, binding pockets)
- Epitopes you want to block or stabilize
- Surface-exposed residues (buried residues are unreachable)

Output Files

RFD3 writes .cif.gz files (compressed mmCIF format) containing backbone atom coordinates (N, CA, C, O) for each generated structure. These files do NOT contain amino acid identities - every position is placeholder (typically glycine or alanine).

5. Stage 2: LigandMPNN / solMPNN - Sequence Optimization

Given a backbone structure from RFD3, this stage solves the inverse folding problem: what amino acid sequence will fold into that shape? The notebook offers two models:

Feature	LigandMPNN	solMPNN
Ligand awareness	Yes	No
Non-standard residues	Supported	Not intended
Best for	Binder design (default)	Soluble redesign
Inference path	ligand_mpnn	protein_mpnn + soluble weights

Both models use the same MPNN (Message Passing Neural Network) architecture but were trained on different datasets. LigandMPNN was trained with ligand-protein complexes and is the default choice for binder design because it understands molecular interfaces. solMPNN was trained to maximize solubility and is useful when you want sequences that are easier to express recombinantly.

Key Parameters

- `designed_chains = ['A']`: Only the binder chain is redesigned; the target sequence stays fixed.
- `number_of_batches = MPNN_SEQS`: How many different sequences to sample per backbone.
- `seed = 42`: For reproducibility across runs.

Output Files

FASTA files (.fa) where each entry is a complete complex sequence (binder + target concatenated). The downstream ColabFold step separates them using the known target length.

6. Stage 3: ColabFold (AF2 Multimer) - Structural Validation

This is the critical validation step. ColabFold uses AlphaFold2 Multimer v3 to predict the 3D structure of each binder-target complex from sequence alone - no template, no prior structure. If the predicted structure closely matches the designed backbone, the design is self-consistent.

MSA Caching Strategy

The notebook uses a smart three-phase approach to minimize API calls:

Phase 1 (one-time): Generate the target MSA via MMseqs2 API and cache it.

Phase 2 (per design): Build 'hybrid paired A3Ms' - the target side gets the full cached MSA while the binder side gets a single sequence (de novo designs have no evolutionary homologs).

Phase 3: Run ColabFold on the .a3m files directly. ColabFold detects the pre-built MSAs and skips all API searches entirely.

This caching approach dramatically speeds up repeated runs. The first run builds the MSA cache (~2-5 min); subsequent runs skip this step.

ColabFold Settings

Parameter	Default	Range	Description
num_models	1	1-5	AF2 model checkpoints to use (more = slower but more diverse)
num_recycle	6	1-20	Recycling iterations (more = better convergence)
use_relax	False	bool	Run Amber relaxation on predicted structures
use_drive_cache	False	bool	Cache MSA in Google Drive (persists across sessions)

The Amber Relax Option (use_relax)

When use_relax = True, ColabFold runs an Amber energy minimization on each predicted structure using OpenMM. This removes steric clashes and improves local geometry. The trade-off is increased runtime (~1-3 min per structure).

IMPORTANT: Amber relaxation requires two extra packages: openmm (the molecular dynamics engine) and pdbfixer (PDB cleaning before minimization). These are installed in cell 0.3. If you see 'ModuleNotFoundError: No module named pdbfixer', re-run cell 0.3.

For initial screening, relaxation is optional. For final candidate selection, molecular dynamics input, or publication figures, enable it.

Confidence Metrics Produced

Metric	Measures	Range	Good Value
ipTM	Interface confidence	0-1	> 0.7
pTM	Overall fold confidence	0-1	> 0.7
pLDDT	Per-residue confidence	0-100	> 80
pAE	Predicted alignment error	0-50 A	< 10 A

7. Stage 4: Screening & Quality Metrics

After ColabFold predicts structures for all designs, this stage extracts quality metrics from the score JSON files, computes binder RMSD (see Section 8), and applies tiered filtering.

Screening Thresholds (Adjustable)

Parameter	Default	Description
ipTM_threshold	0.5	Minimum interface predicted TM-score
pLDDT_threshold	80	Minimum average pLDDT
RMSD_threshold	2.0 Å	Maximum binder RMSD (design vs prediction)

A design passes the filter only if ALL three conditions are met simultaneously. The thresholds are exposed as Colab form sliders so you can adjust them without editing code.

Tiered Quality Assessment

Tier	Criteria	Use Case
Tier 1 (green)	ipTM > 0.4	Workshop / exploratory
Tier 2 (blue)	ipTM > 0.7, pTM > 0.7, pLDDT > 80	Experimental testing
Tier 3 (red)	RMSD < 2.0, ipTM > 0.8, pLDDT > 85	Publication quality

Output: design_metrics.csv

The screening cell produces a CSV file with one row per design. Columns include:

- design_id, sequence, length
- binder_rmsd, binder_ca_atoms (RMSD between RFD3 backbone and AF2 prediction)
- ipTM, pTM, pLDDT, pAE_interface
- rfd3_backbone_file, cf_pdb_file (matched file pairs)
- pass_filter (boolean)

This CSV is the main deliverable for downstream analysis.

8. Binder RMSD: RFDiffusion vs ColabFold

One of the most informative metrics is the backbone RMSD (Root Mean Square Deviation) between the binder as designed by RFDiffusion3 and the binder as predicted by ColabFold. A low RMSD means the sequence folds back into the intended shape.

How It Works

1. For each design, the notebook matches the RFD3 backbone file (.cif.gz) with the corresponding ColabFold prediction (.pdb).
2. It extracts the binder chain (chain A) C-alpha atoms from both structures.
3. Using BioPython's Superimposer, it aligns the two sets of atoms and computes the RMSD after optimal superposition.
4. The RMSD value (in Angstroms) is added to the metrics table.

Interpreting RMSD Values

RMSD (Å)	Interpretation	Confidence
< 1.0	Excellent - near-identical fold	Very high
1.0 - 2.0	Good - minor deviations	High
2.0 - 4.0	Fair - noticeable differences	Moderate
> 4.0	Poor - significantly different fold	Low

A design with high ipTM but high RMSD may indicate that ColabFold found a confident alternative fold that differs from the design intent. These should be inspected visually in the 3D viewer.

9. Quality Metrics Plots

The notebook generates a 2x3 panel of plots (saved as a PNG image) that visualize the distribution of quality metrics across all designs. All dashed threshold lines are pulled from the screening cell sliders.

Panel Layout

Position	Plot Type	X-axis	Color / Notes
Top-left	Histogram	ipTM	Red dashed line at ipTM_threshold
Top-center	Histogram	pTM	No threshold line
Top-right	Histogram	pLDDT	Red dashed line at pLDDT_threshold
Bottom-left	Histogram	Binder RMSD	Red dashed line at RMSD_threshold
Bottom-center	Scatter	ipTM vs pLDDT	Color = binder RMSD (plasma_r)
Bottom-right	Scatter	RMSD vs ipTM	Color = pLDDT (viridis)

The plot is saved automatically to the results directory as <TARGET_NAME>_quality_metrics.png and included in the final ZIP download.

10. 3D Visualization with py3Dmol

The notebook provides four interactive 3D viewers using py3Dmol. These render directly in the Colab output area.

6.1 - Visualize Target Protein

Shows the target protein in cartoon representation with hotspot residues highlighted as red spheres. Use this to verify your PDB loaded correctly and hotspots are in sensible locations.

6.2 - Visualize Best Design

Shows the highest-scoring design (by ipTM). The binder is colored in rainbow (N-terminus blue to C-terminus red) and the target is in light blue.

6.3 - Compare Designs (Grid)

Side-by-side grid of top N designs (adjustable slider, default 4). Useful for quickly spotting structural diversity among top candidates.

6.4 - pLDDT Confidence Coloring

Colors each residue by AlphaFold2 confidence (stored in B-factor):

- Blue (> 90): Very high confidence - reliable
- Cyan (70-90): High confidence - mostly reliable
- Yellow (50-70): Low confidence - uncertain
- Orange (< 50): Very low confidence - likely disordered

Controls

- Rotate: Left-click and drag
- Zoom: Scroll wheel
- Pan: Right-click and drag

11. Downloading Results

The final cell packages all outputs into a ZIP file and triggers a browser download. The ZIP structure is:

```
<TARGET_NAME>_designs.zip
  design_metrics.csv
  01_backbones/      (RFD3 .cif.gz files)
  02_sequences/      (MPNN .fasta files)
  03_predictions/    (ColabFold .pdb files)
  04_plots/          (quality metrics .png images)
```

12. Parameter Reference

Quick reference for all adjustable parameters across the notebook.

Design Parameters (Cell 1.0)

Parameter	Default	Suggested Range
TARGET_NAME	FPR2_v2	Any short identifier
TARGET_CHAIN	B	A-F (must match your PDB)
BINDER_MIN	5	5-200 residues
BINDER_MAX	20	15-250 residues
NUM_BACKBONES	4	4-100
MPNN_SEQS	4	1-32
MPNN_MODEL	ligandmpnn	ligandmpnn or solmpnn

ColabFold Parameters (Cell 4.0)

Parameter	Default	When to Change
num_models	1	Increase to 3-5 for final candidates
num_recycle	6	Increase to 12+ for difficult targets
use_relax	False	True for publication / MD input (needs openmm + pdbfixer)
use_drive_cache	False	True to persist MSA across sessions

Screening Parameters (Cell 5.0)

Parameter	Default	Workshop Tier	Pub. Quality Tier
ipTM_threshold	0.5	> 0.4	> 0.8
pLDDT_threshold	80	> 60	> 85
RMSD_threshold	2.0 Å	< 5.0 Å	< 2.0 Å

13. Troubleshooting

No GPU detected

Go to Runtime > Change runtime type > GPU (T4 or A100). Then Runtime > Restart and run all.

JAX / DropVar error

Re-run cell 0.3b which pins JAX < 0.11 for ColabFold compatibility.

ModuleNotFoundError: pdbfixer (when use_relax = True)

ColabFold's Amber relax requires openmm and pdbfixer. Re-run cell 0.3 which installs both packages. If the error persists, run manually:

```
!pip install openmm pdbfixer
```

ColabFold hangs at MMseqs2

The MMseqs2 API can be slow during peak hours. The MSA caching cell (4.0) avoids this on subsequent runs. Wait or retry later.

No MPNN sequences generated

Check that rfd3 produced outputs first. Run '!mpnn --help' to verify the CLI flags match your Foundry version.

RMSD is NaN

The matching between RFD3 backbone and ColabFold PDB failed - usually because the design_id naming differs. Check that the binder chain is 'A' in both structures.

Checkpoint not found

Re-run cell 1.2. If 'foundry install' fails, check your internet connection. Foundry will also search ~/.foundry/checkpoints.

Out of GPU memory

Reduce binder length (BINDER_MAX), or switch to an A100 runtime. ColabFold with num_models > 1 is the main memory consumer.

solMPNN: 'unknown model_type'

solMPNN uses the protein_mpnn inference path with soluble-trained weights. The notebook handles this mapping automatically. Ensure cell 1.2 installed the solublempnn checkpoint.

14. Next Steps & References

What to Do After This Workshop

1. Scale up: Increase NUM_BACKBONES to 20-50 and MPNN_SEQS to 8-16 for a more thorough exploration of design space.
2. Prioritize candidates: Sort by low RMSD + high ipTM. Designs where both metrics are strong are the most promising.
3. Molecular dynamics: Run short MD simulations (e.g. OpenMM, GROMACS) on top candidates to assess structural stability.
4. Experimental validation: Order gene synthesis for 5-10 top designs. Test binding with SPR, BLI, or pull-down assays.
5. Iterate: Use experimental feedback to refine hotspots, binder length, and sequence constraints for a second round of design.

Key References

- [1] Watson et al. (2023). De novo design of protein structure and function with RFdiffusion. *Nature*, 620, 1089-1100.
- [2] Dauparas et al. (2022). Robust deep learning-based protein sequence design using ProteinMPNN. *Science*, 378(6615), 49-56.
- [3] Mirdita et al. (2022). ColabFold: making protein folding accessible to all. *Nature Methods*, 19(6), 679-682.
- [4] Krishna et al. (2024). Generalized biomolecular modeling and design with RoseTTAFold All-Atom (RFdiffusion3). *bioRxiv*.
- [5] Jumper et al. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature*, 596, 583-589.

Useful Links

- Foundry (rc-foundry): <https://pypi.org/project/rc-foundry/>
- ColabFold GitHub: <https://github.com/sokrypton/ColabFold>
- RFdiffusion paper: <https://doi.org/10.1038/s41586-023-06415-8>
- ProteinMPNN paper: <https://doi.org/10.1126/science.add2187>
- PDB database: <https://www.rcsb.org/>